DraftRun Pipeline AS IS

Current as of Slice 2.17.4.6.0.3.4: Validation and Revision Loop Runtime Guard.

This document is the maintained technical map of the current DraftRun generation pipeline. It describes the running system as it exists now, not the target design. Update this file and regenerate the PDF whenever a slice changes DraftRun steps, artifacts, role model usage, context flow, retry/fallback behavior, validation, ranking, revision, or trace semantics.

PDF quick-view copy: `docs/architecture/DRAFT_RUN_PIPELINE_AS_IS.pdf`.

Regenerate PDF:

```
python scripts/generate-draft-run-pipeline-pdf.py
```

1. Core concepts

Concept	Meaning	Stored where
`DraftRun`	Parent orchestration run for one approved post fabula. It owns step order, status, final draft, and child `AiRun` ids.	`var/glavred-draft-runs.sqlite3`
`DraftRunStep`	One logical pipeline stage: context, source intent, public evidence, feasibility, contract, rules, planning, candidates, validation, complete.	`draft_run_steps.artifact_payload`
`AiRun`	One model/provider call or audited fallback inside a DraftRun.	`var/glavred-ai-runs.sqlite3`
`SourceLedger`	Claim/provenance inventory with allowed use, confidence, warnings, risks, and forbidden inferences.	`context` and enriched `publicEvidence` artifacts
`EvidenceInterpretation`	Editorial interpretation of accepted evidence: implications, tensions, usable examples, limits, forbidden overclaims, reader-value hooks, and rejected uses.	`rulePack.evidenceInterpretation`, child `AiRun` trace
`PostContract`	Locked editorial intent: thesis, audience, CTA, allowed claims, forbidden moves, size contract, topic/fabula obligations.	`postContract` artifact
`RuleRegistrySnapshot`	Machine-readable rules and validator bindings derived from contract, ledger, topic, fabula, and publisher rules.	`rulePack.ruleRegistrySnapshot`
`ArticleDossier`	DraftRun-local article memory built from artifacts. It is not workspace persistence and not vector storage.	selected step artifacts
`ContextPack`	Role-specific subset of ArticleDossier passed to LLM calls.	step artifacts and child `AiRun.requestPayload`
`EditorialCritiqueReport`	Report-only prosecutor/editor critique of candidate idea strength, tension, author stance, source integration, and reader value.	`validation.editorialCritiqueReport`, child `AiRun` trace
`AlternativeAngleTournament`	Critic-driven challenger route and candidate used to escape a weak local optimum.	`validation.alternativeAngleTournament`, child `AiRun` trace

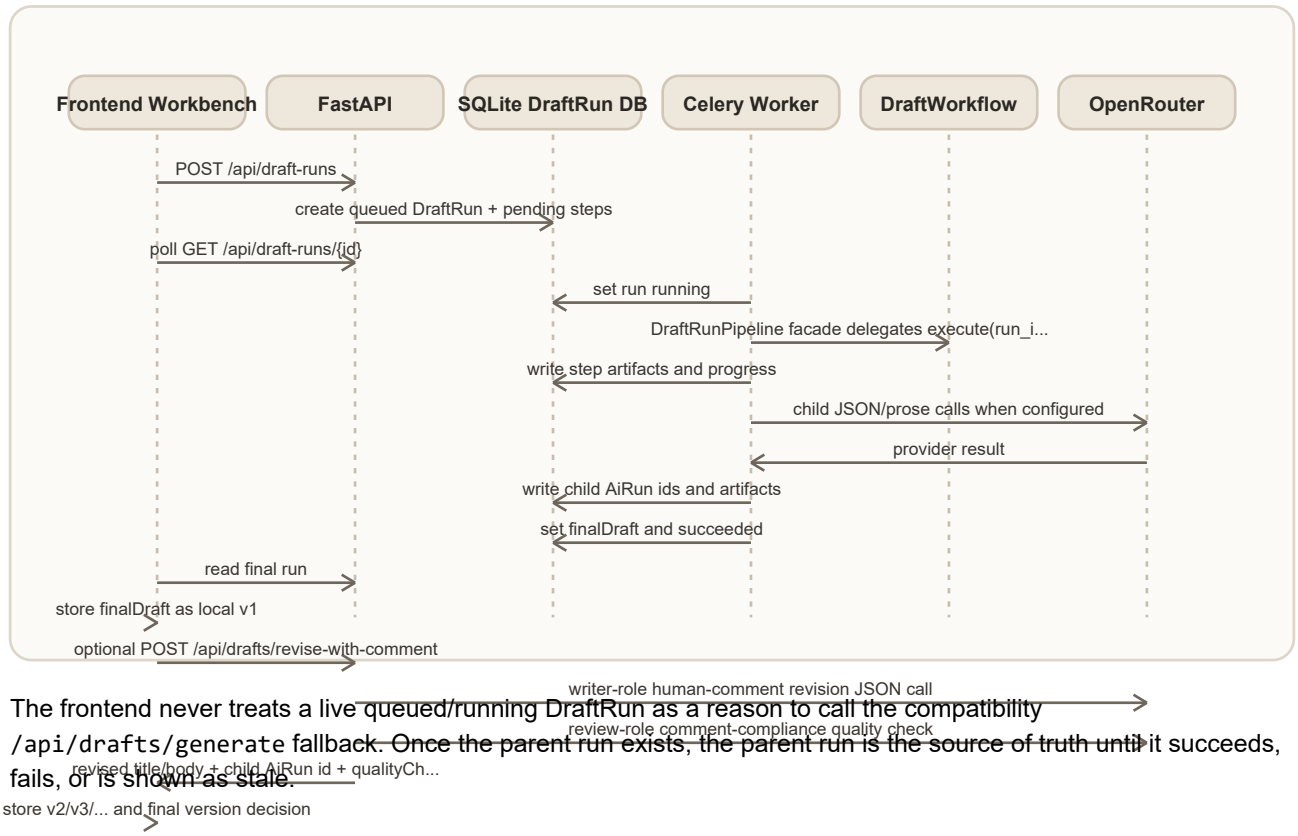
Concept	Meaning	Stored where
`RevisionLoop`	Bounded editorial optimization loop: validator repair goals plus editorial goals, old-vs-new dimension scoring, accepted/rejected revisions, and rejected moves.	`validation.rankingRevision.revisionLoop`
`FinalQualityGate`	Last machine acceptance layer for the delivered final draft as public prose; combines deterministic checks with an independent final-gate model review and may run bounded writer repair cycles if actionable gate findings require it. It separates real missing attribution from diagnostic attribution handoff noise.	`validation.rankingRevision.finalQualityGate`
`DraftRunBudget`	Effective research/execution caps derived from `Fabula.researchDepth` and backend execution mode.	`context.draftRunBudget`, downstream budget metadata
`PayloadBudget`	Provider-input boundary for LLM calls: operation profile, execution mode, prompt/token estimates, sent/trimmed counts, suppressed fields, semantic inputs, quality risk, and over-budget incidents.	child `AiRun.requestPayload.payloadBudget`, attempts, `operationEnvelope.payloadStats`
`ValidationRuntimeBudget`	Runtime boundary for the validation/revision loop: wall-clock, LLM-call, revision-cycle, pairwise-round, final-repair, non-improvement, heartbeat, slow-but-healthy, and canonical stop-reason accounting.	`validation.progress.runtimeBudget`, `validation.rankingRevision.runtimeBudget`, `validation.rankingRevision.revisionLoop.runtimeBudget`
`finalDraft`	The selected draft returned to the frontend after validation, ranking, revision loop, and final quality gate.	parent DraftRun
`DraftVersion`	Immutable editor-facing version of the delivered draft. `v1` is the machine final draft; later versions come from human-comment revisions or manual edits.	local workspace `postDraft.versions[]`
`HumanCommentRevisionQualityCheck`	Diagnostic review of one human-comment revision: comment compliance, source-marker preservation, public-prose health, internal jargon leaks, and base-version regression risks. It never blocks saving the version.	local workspace `postDraft.versions[].qualityCheck`, child `AiRun` trace
`EditorDecisionSnapshot`	Human final-selection record linking the chosen version to machine trace summaries, unresolved risks, comments, and manual edit counts.	local workspace `finalText.editorDecisionSnapshot`
`EditorialLearningAuthorNote`	Auto-created author-memory note summarizing what the editor appears to have taught the system through final version choice, comments, manual edits, rejected versions, and quality checks. Starts as `pendingReview`; only accepted notes influence author-position inference.	local workspace `authorNotes[]`, type `editorialLearning`

2. Runtime topology

Runtime entrypoints are compatibility-preserving:

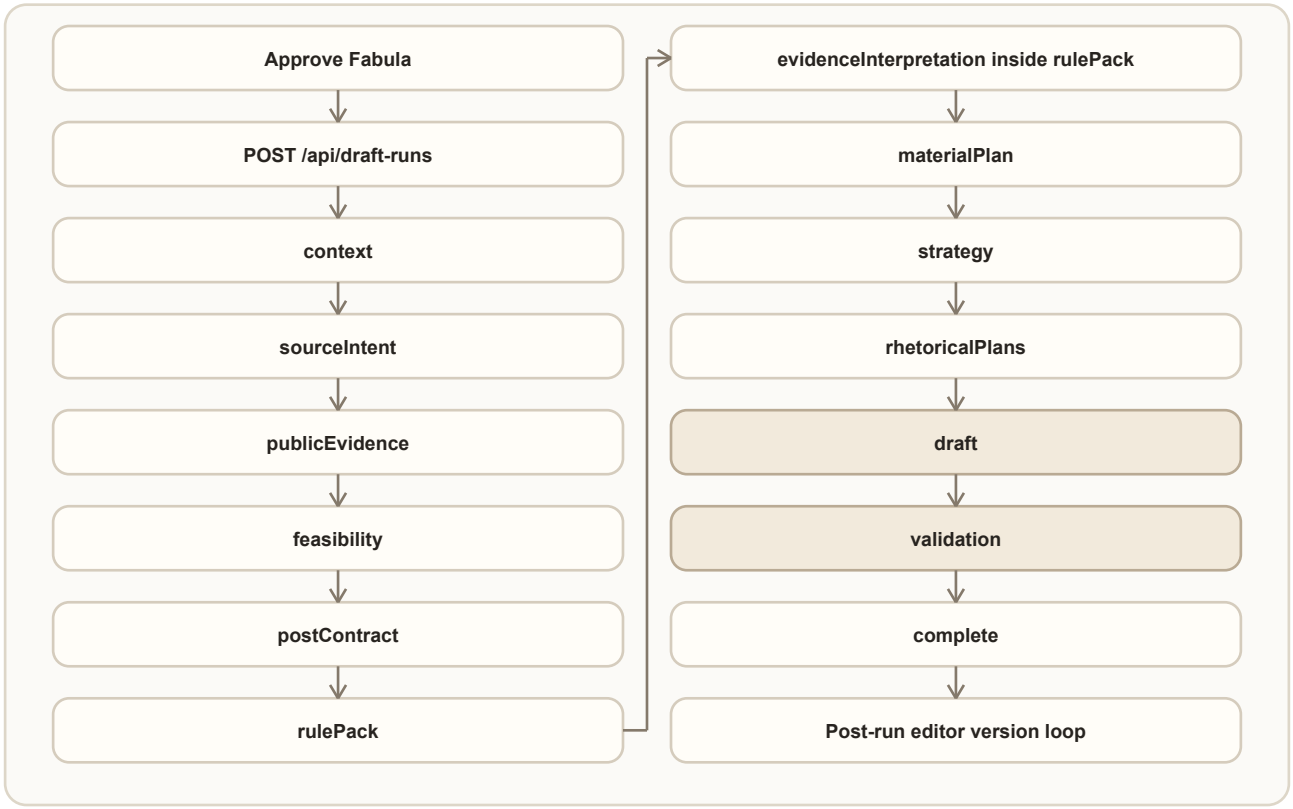
- the Celery/API path still enters the drafting runtime through `backend.app.application.draft_run_pipeline.DraftRunPipeline`;

- `DraftRunPipeline` is now a thin facade that keeps the previous constructor and `execute(run_id)` method;
- sequencing is delegated to `backend.app.drafting.application.workflow.DraftWorkflow` through an ordered `DraftStepRegistry`;
- individual step services, prompts, provider calls, artifact payloads, progress semantics, child `AiRun` persistence, and blocked/failure behavior are unchanged.



The frontend never treats a live queued/running DraftRun as a reason to call the compatibility `/api/drafts/generate` fallback. Once the parent run exists, the parent run is the source of truth until it succeeds, fails, or is shown as stale.

3. Current step order



The backend enum is:

```
context -> sourceIntent -> publicEvidence -> feasibility -> postContract -> rulePack -> materialPlan -> strategy -> rhetoricalPlans -> draft -> validation -> complete
```

The ordered runtime registry uses the same sequence. It is a package-boundary refactor, not a semantic step change.

4. Step-by-step technical flow

The role system is not a separate chat room. Roles interact through persisted DraftRun artifacts. Each LLM call chooses a role model through `DraftModelRoleResolver`, receives the artifact set and the role-specific `ContextPack`, writes a child `AiRun`, and stores its result back into the parent step artifact. The next role reads that artifact instead of relying on hidden conversation state.

Role-aware execution map:

Step	Active role	Model selection	Context passed to the role	Output consumed by
`context`	none	no model	frontend snapshot only	source intent, ledger, feasibility, DraftRun budget
`sourceIntent`	`research`	`DRAFT_RESEARCH_MODEL`, then default	brief sources, context summary, initial ledger, DraftRun budget	public evidence
`publicEvidence` search	web search	`OPENROUTER_WEB_SEARCH_MODEL`	budget-capped research tasks and built search query	evidence synthesis
`publicEvidence` synthesis	`research`	`DRAFT_RESEARCH_MODEL`, then default	accepted public evidence, initial ledger, context	enriched ledger, dossier
`feasibility`	none	no model	enriched ledger and warnings	contract or blocked completion

Step	Active role	Model selection	Context passed to the role	Output consumed by
`postContract`	none	no model	brief, ledger, feasibility, size settings	rules, planning, validation
`rulePack`	none	no model	contract, ledger, publisher/topic/fabula rules	evidence interpretation, validation, revision
`evidenceInterpretation` inside `rulePack`	`strategy`	`DRAFT_STRATEGY_MODEL`, then default, repair, backup	compact enriched ledger, evidence synthesis, accepted public evidence, contract, relevant rule subset, strategy `ContextPack`, timeout envelope	material plan, dossier, context packs
`materialPlan`	`strategy`	`DRAFT_STRATEGY_MODEL`, then default, repair, backup	strategy `ContextPack`, usable evidence candidates, evidence interpretation, contract, registry	draft strategy
`strategy`	`strategy`	`DRAFT_STRATEGY_MODEL`, then default	material plan, rules, contract, strategy `ContextPack`	rhetorical plans
`rhetoricalPlans`	`strategy`	`DRAFT_STRATEGY_MODEL`, then default, repair, backup	strategy, material plan, claim/rule ids	writer candidates
`draft`	`writer`	`DRAFT_WRITER_MODEL`, then default, repair, backup, domain-safe fallback	writer `ContextPack`, one rhetorical plan, material evidence, contract, writer generation params	validation and ranking
`validation` lint	none	no model	candidates, contract, registry, ledger	LLM validation and ranking
`validation` LLM review	`review`	`DRAFT_REVIEW_MODEL`, then default, repair, backup	review `ContextPack`, candidates, deterministic findings	pairwise ranking
`validation` editorial critique	`critic`	`DRAFT_CRITIC_MODEL`, then default, repair, backup	critic `ContextPack`, candidates, evidence interpretation, validation findings	trace, dossier, future critique-aware ranking
`validation` alternative angle	`anotherAngle` then `writer`	`DRAFT_ANOTHER_ANGLE_MODEL` for route, `DRAFT_WRITER_MODEL` for challenger prose, repair, backup, no fake challenger fallback	initial validation, editorial critique, another-angle `ContextPack`, writer `ContextPack`, another-angle/writer generation params	final validation and ranking
`validation` ranking	`review`	`DRAFT_REVIEW_MODEL`, then default, repair, backup	merged candidates, old scorecard, deterministic and LLM findings	revision loop
`validation` revision	`writer`	`DRAFT_WRITER_MODEL`, then default, repair, backup, failed/not-run if no usable JSON	current best candidate, repair goals, anti-regression constraints, writer `ContextPack`, revision generation params	regression guard and final draft
`validation` final quality gate	`finalGate`, then optional `writer`	`DRAFT_FINAL_GATE_MODEL`, or independent critic/review/default fallback; `DRAFT_WRITER_MODEL` for bounded final repair cycles	delivered final candidate, `FinalQualityContract`, validation reports, critique, evidence interpretation, ledger, contract, rule registry, material plan	accepted public draft or rejected repair trace
`complete`	none	no model	final DraftRun state	frontend

Step	Active role	Model selection	Context passed to the role	Output consumed by
post-run human comment revision	`writer`	`DRAFT_WRITER_MODEL`, then default, repair, backup, failed if no usable JSON	selected `DraftVersion`, editor comment, compact machine trace context from final quality gate, revision loop, alternative angle, validation, contract/material evidence	new local `DraftVersion`; not a new DraftRun
post-run human revision quality check	`review`	`DRAFT_REVIEW_MODEL`, then default, repair, backup, `notRun` if no usable JSON	base version, revised version, editor comment, compact machine trace context, source markers and public-prose constraints	`DraftVersion.qualityCheck`; child `AiRun`, not a DraftRun step

Practical reading rule: when a step says `Role/model handoff: strategy`, it does not mean that the strategy model talks directly to the writer model. It means the strategy role writes a structured artifact, then the writer role later receives that artifact plus its own compact context pack.

4.1 `context`

Purpose: build the normalized local context for the selected work item.

Input:

- approved `PostBrief`;
- `EditorialModel`;
- `draftContext` snapshot from frontend: work item, plan slot, candidate, signal, topic, fabula, publisher rules, author-position evidence, publication-size context.

Processing:

- `build_draft_run_context_summary(...)` normalizes the frontend snapshot;
- `SourceLedgerBuilder` builds the initial internal claim ledger.
- `DraftRunBudgetResolver` combines `fabula.researchDepth` with `DRAFT_RUN_EXECUTION_MODE`.

Output:

- `contextSummary`;
- initial `sourceLedger`;
- `draftRunBudget`;
- `missingContext` and compatibility metadata when links are absent.

Role/model handoff:

- no LLM role is used;
- this step creates the first artifact boundary for later roles;
- downstream `research`, `strategy`, `writer`, and `review` calls never read the raw frontend workspace directly after this point.

Trace: `steps[].stepKey = context`.

4.2 sourceIntent

Purpose: convert approved brief sources into an explicit research plan.

Input:

- PostBrief.sources;
- context summary;
- initial SourceLedger.

Processing:

- SourceIntentNormalizer classifies URLs, named sources, human-language requests, proof needs, framing hints, exclusions, and unknown lines;
- ResearchPlanService may call OpenRouter through the research role;
- deterministic fallback preserves original wording when provider planning fails;
- source_research_budgeting caps executable verification tasks and records budgetTrace.budgetSkipped instead of silently dropping them.

Output:

- normalized sourceIntent;
- researchPlan;
- draftRunBudget and budgetTrace;
- child AiRun when OpenRouter is used.

Role/model handoff:

- active role: research;
- model resolver chooses DRAFT_RESEARCH_MODEL or falls back to OPENROUTER_DEFAULT_MODEL;
- the role receives sources, context summary, and the initial ledger;
- it returns a ResearchPlan, not proof;
- publicEvidence consumes the plan and decides what can actually be read or searched.

Trace: sourceIntent step and child AiRun.requestPayload.draftRunStep = sourceIntentResearchPlan.

4.3 publicEvidence

Purpose: execute available public evidence tasks and enrich the ledger.

Input:

- sourceIntent artifact;
- researchPlan;
- current context artifact.

Processing:

- exact URL tasks are read by the URL reader;
- public search tasks call OpenRouter web search only when web tools are enabled;
- retrieval tasks, URL reads, and search result counts are capped by `DraftRunBudget`;
- disabled or unavailable search tasks remain explicit attempts, not proof;
- relevance guard rejects search-result drift;
- `EvidenceSynthesis` reconciles accepted public evidence into external claims;
- `SourceLedgerExternalEvidenceMerger` merges accepted external claims into the ledger;
- accepted evidence items and external ledger claims are trimmed by budget before downstream planning;
- operation progress is persisted during URL/search/synthesis work.

Output:

- `PublicEvidenceBatch`: attempts, accepted evidence items, warnings;
- `EvidenceSynthesis`: external claims, decisions, warnings;
- enriched `SourceLedger`;
- budget metadata: used counts, cap hits, skipped tasks, trimmed evidence/claims;
- initial `ArticleDossier` and all role `ContextPacks`.

Role/model handoff:

- public search uses the web-search model/config;
- evidence synthesis uses `research`.
- web search receives concrete URL/search tasks and built search queries;
- `research` synthesis receives accepted evidence candidates and decides how they should become external ledger claims;
- downstream roles consume the enriched ledger and dossier, not raw search output.

Trace:

- `publicEvidence` artifact;
- child `AiRun.requestPayload.draftRunStep = externalEvidenceSynthesis`;
- child web-search `AiRun` records when search is enabled.

AS IS limitation:

- public evidence retrieval and synthesis do not write prose directly. Accepted evidence is merged into the ledger here, then interpreted inside `rulePack` before material planning and writing.

4.4 `feasibility`

Purpose: decide whether it is safe to generate prose.

Input:

- enriched SourceLedger;
- context warnings;

- evidence availability.

Processing:

- deterministic quality gate classifies the run as feasible, feasible with constraints, needs research, needs human decision, or infeasible.

Output:

- `FeasibilityReport`;
- if blocked: `complete.status = blocked`, `finalDraft = null`, no local fallback.

Role/model handoff:

- no LLM role is used;
- this deterministic gate decides whether later roles are allowed to write prose;
- if blocked, `strategy`, `writer`, and `review` are not called.

Trace: `feasibility` step.

4.5 `postContract`

Purpose: lock what the post must preserve.

Input:

- approved brief;
- context summary;
- enriched SourceLedger;
- feasibility result;
- publication-size profile and fabula size intent.

Processing:

- `PostContractBuilder` resolves title, thesis, audience, value, goal, CTA, platform/date/time, topic/fabula ids, allowed claim ids, qualified claims, forbidden moves, evidence obligations, fabula obligations, risks, and `PublicationSizeContract`.

Output:

- provider-free `PostContract`.

Role/model handoff:

- no LLM role is used;
- this step creates the invariant contract that all later roles must preserve;
- `strategy` may plan within the contract, `writer` may execute it, and `review` evaluates candidates against it.

Trace: `postContract` step.

4.6 `rulePack`

Purpose: turn the contract and editorial rules into machine-readable rules.

Input:

- enriched context artifact;
- SourceLedger;
- FeasibilityReport;
- PostContract;
- topic/fabula/publisher rules.

Processing:

- `rulePack` is marked `running` before provider-backed evidence interpretation, so a

long JSON retry path is visible as `rulePack`, not as stale `postContract`;

- `RuleRegistrySnapshot` is compiled first;
- compatibility `RulePack` is derived from the registry;
- the provider request is compacted before JSON generation: full `ruleRegistrySnapshot`

stays in the artifact, while the model receives relevant hard/critical/evidence/ style/attribution/topic/fabula rules, accepted public evidence summaries, external ledger claims, evidence synthesis, and the locked contract;

- each provider attempt runs inside `DRAFT_EVIDENCE_INTERPRETATION_TIMEOUT_SECONDS`

so a stuck strategy-model call can fail, write a child `AiRun`, and continue to repair/backup/fallback instead of leaving `rulePack` running forever;

- `EvidenceInterpretationService` converts accepted internal/external evidence into

editorial implications, tensions, examples, limits, forbidden overclaims, reader value hooks, and rejected evidence uses before material planning.

Output:

- `ruleRegistrySnapshot`;
- compatibility rule-pack fields;
- `evidenceInterpretation`;
- `attempts[]` for the interpretation provider/repair/backup path;
- updated `ArticleDossier` and `ContextPacks`.

Role/model handoff:

- rule compilation itself uses no LLM role;
- evidence interpretation uses the `strategy` role through `DRAFT_STRATEGY_MODEL`,

then default, repair, optional backup, and finally deterministic fallback;

- this step translates editorial obligations into machine-readable rule ids and then

translates evidence into usable editorial meaning;

- later role prompts receive rule ids and validator bindings so they do not work

from anonymous prose constraints only;

- material planning and writing receive interpretation artifacts instead of raw

citation snippets as their main source context.

Trace:

- `rulePack` step;
- child `AiRun.requestPayload.draftRunStep = evidenceInterpretation` when provider interpretation runs;
- `artifactPayload.progress.operations[]` contains `evidenceInterpretation` primary/repair/backup attempts while the step is running;
- child `AiRun.requestPayload.inputStats` records original/compact rule counts, claim counts, evidence counts, prompt char estimate, selected model, and timeout seconds;
- attempt records may include `status=timeout` plus operation timing notes such as prompt build, provider request, JSON/shape validation, and AiRun persistence;
- `rulePack.evidenceInterpretation` readable section in `/ai-runs?runId=...`.

4.7 `materialPlan`

Purpose: decide what material is usable before strategy and writing.

Input:

- context summary;
- RulePack and RuleRegistry;
- EvidenceInterpretation;
- enriched SourceLedger;
- PostContract;
- strategy `ContextPack`.

Processing:

- material planner receives `usableEvidenceCandidates`;
- the candidate list is capped by `DraftRunBudget.maxUsableEvidenceCandidates`;
- material planner receives `evidenceInterpretation` and should prefer implications, usable examples, limits, and forbidden overclaims over raw public snippets;
- it must either choose usable evidence or explain rejection reasons;
- accountability guard rejects empty/unexplained evidence selection;
- retry sequence: primary -> repair -> optional backup model -> emergency deterministic fallback.

Output:

- `MaterialPlan`;
- `availableEvidence`;
- `rejectedEvidence`;
- `claimsRequiringAttribution`;
- `qualifiedClaims`;
- `attempts[]`;

- accountability metadata;
- updated `ArticleDossier` and `ContextPacks`.

Role/model handoff:

- active role: `strategy`;
- model resolver chooses `DRAFT_STRATEGY_MODEL`, then default, then repair/backup where applicable;
- the role receives the strategy `ContextPack`, usable evidence candidates, ledger claim ids, contract, and registry;
- it must return selected or rejected evidence with reasons;
- `strategy` and `rhetoricalPlans` consume the material plan instead of scanning the whole ledger again.

Trace:

- `materialPlan` step;
- child `AiRun.requestPayload.draftRunStep = materialPlan`.

4.8 `strategy`

Purpose: define the main strategy for the post.

Input:

- context summary;
- RulePack;
- MaterialPlan;
- strategy `ContextPack`.

Processing:

- OpenRouter JSON call or deterministic fallback builds the draft strategy.

Output:

- `DraftStrategy`: thesis angle, opening move, argument sequence, fabula usage, CTA plan, forbidden moves, tone notes;
- updated `ArticleDossier` and `ContextPacks`.

Role/model handoff:

- active role: `strategy`;
- model resolver chooses `DRAFT_STRATEGY_MODEL` or default;
- the role receives MaterialPlan plus the strategy `ContextPack`;
- it creates the main route of argument, but cannot change the PostContract;
- `rhetoricalPlans` expands this strategy into several executable routes.

Trace: `strategy` step and child `AiRun.requestPayload.draftRunStep = strategy`.

4.9 `rhetoricalPlans`

Purpose: build several routes for executing the same contract.

Input:

- context summary;
- RulePack;
- MaterialPlan;
- DraftStrategy.

Processing:

- one JSON call returns 2-3 rhetorical plans;
- retry sequence: primary -> repair -> optional backup -> deterministic fallback.

Output:

- `RhetoricalPlanSet`;
- plan moves, claim ids to use/avoid, required rule ids, CTA route, risks;
- updated `ArticleDossier` and `ContextPacks`.

Role/model handoff:

- active role: `strategy`;
- model resolver chooses `DRAFT_STRATEGY_MODEL`, then repair/backup if JSON shape is invalid;
- the role receives contract, material plan, draft strategy, claim ids, rule ids, and the strategy `ContextPack`;
- it returns alternative rhetorical plans that the writer must execute;
- the writer must not invent a new route outside these plan artifacts.

Trace:

- `rhetoricalPlans` step;
- child `AiRun.requestPayload.draftRunStep = rhetoricalPlans`.

4.10 `draft`

Purpose: generate candidate drafts from rhetorical plans.

Input:

- request brief;
- context summary;
- RulePack;
- MaterialPlan;
- DraftStrategy;
- RhetoricalPlanSet;

- EvidenceInterpretation;
- writer `ContextPack`.

Processing:

- one candidate is generated per rhetorical plan;
- the executed candidate count is capped by `DraftRunBudget.maxDraftCandidates` before provider calls are attempted;
- each provider failure falls back per candidate, not for the whole step;
- candidate publishability guard marks candidates eligible, penalized, or excluded;
- deterministic v1 selection creates an initial scorecard.

Output:

- `candidates[]`;
- `selection`;
- `scorecard`;
- child `AiRun` ids;
- updated `ArticleDossier` and `ContextPacks`.

Role/model handoff:

- active role: `writer`;
- model resolver chooses `DRAFT_WRITER_MODEL` or default;
- each candidate call receives one rhetorical plan, writer `ContextPack`, material evidence, evidence interpretation, allowed claim ids, forbidden moves, and size contract;
- each JSON candidate attempt follows the universal retry sequence: primary writer model, repair prompt, optional backup model, then domain-safe deterministic fallback only if provider attempts are exhausted;
- primary writer attempts use `DRAFT_WRITER_TEMPERATURE` and `DRAFT_WRITER_TOP_P`;
- repair and backup attempts use `DRAFT_JSON_REPAIR_TEMPERATURE`, not the creative writer temperature;
- writer prompts forbid leaking internal artifact names such as `SourceLedger`, `publicEvidence`, `validators`, `RuleRegistry`, or `PostContract` as unexplained public dev-jargon;
- candidates write prose, but selection still remains downstream responsibility;
- `review` receives all candidates plus validation context.

Trace:

- `draft` step;
- child `AiRun.requestPayload.draftRunStep = draftCandidate`.

AS IS behavior:

- writer prompts now receive interpreted implications, examples, limits, and forbidden overclaims, so public evidence should shape the argument instead of being pasted as a decorative citation block;
- each provider attempt is visible through child `AiRun` audit with model role,

selected model, attempt label, repair/backup marker, generation params, and safe validation error.

4.11 `validation`

Purpose: validate candidates, rank them, revise the selected candidate, and choose the final draft.

Input:

- draft artifact;
- context artifact;
- RulePack and RuleRegistry;
- MaterialPlan;
- review/writer context packs through child services.

Processing:

1. deterministic lint checks all candidates;
2. LLM validation checks all candidates and separates actionable findings from observations;
3. editorial critique challenges all candidates for idea strength, tension, author

stance, source integration, generic prose, and reader value;

4. alternative-angle tournament may add one critic-driven challenger candidate; both

route JSON and challenger-prose JSON use primary/repair/backup attempts, and no fake deterministic challenger is invented when all attempts fail;

5. final validation report is extended when a challenger enters the pool: initial

candidate reports are reused, provider-heavy validation runs only for the new challenger, and the reports are merged into one final candidate-pool report;

6. pairwise ranking chooses the best candidate from the merged candidate pool;
7. deep revision loop builds validator and editorial goals, revises, re-runs

deterministic validation, compares previous best vs revised across editorial dimensions, and accepts only targeted improvement without regression. Accepted cycles record concrete reasons such as resolved goals, clear pairwise win, and no deterministic regression;

8. final quality gate builds a `FinalQualityContract` from Fabula research depth,

publication kind, post contract, rule registry, and material plan. It checks the delivered candidate with deterministic public-prose rules and an independent `finalGate` model review. If it leaks internal pipeline names, reads like a source dump, loses author stance relative to the contract, or misses reader value, the gate creates bounded final repair instructions for the writer. The number of final repair cycles is controlled by `DRAFT_FINAL_REPAIR_MAX_ITERATIONS` (default `2`). A repaired draft is accepted only when deterministic regression and gate findings improve. Attribution warnings are actionable only when they resolve to source-backed claim ids with expected markers that are missing from the final body. Free-text attribution requirements that cannot be resolved become diagnostic handoff noise; if the independent final-gate review passes source integration, that diagnostic noise does not launch final repair;

9. provider-heavy validation sub-operations persist progress as they run; if a late

operation fails, the operation is marked `failed`, safe error details are saved, and the previous best candidate is kept when one exists.

Output:

- deterministic validation report;

- `llmValidationReport`;

- `editorialCritiqueReport`;

- `alternativeAngleTournament`;

- `rankingRevision`;

- `revisionLoop` with editorial goals, dimension scores, rejected moves, and stop reason;

- `finalQualityGate` with final quality contract, independent review attempts, final-draft status, public-prose status, source-dump risk, actionable attribution findings, diagnostic attribution noise, attribution review closure, final repair goals, repair cycles, repair result, and gate-level final decision;

- `progress` with nested validation operations, child `AiRun` ids, failed-operation status, and safe error details when a provider-heavy sub-operation fails;

- final draft candidate decision;

- updated `ArticleDossier` and `ContextPacks`.

Role/model handoff:

- `review` for LLM validation and pairwise ranking;

- `critic` for report-only editorial critique;

- `anotherAngle` for challenger route generation;

- `finalGate` for independent final public-prose acceptance;

- `writer` for directed revision and final public-prose repair.

- deterministic lint runs first without a model;

- `review` receives candidates, review `ContextPack`, deterministic findings, rules, ledger, and contract, then returns report-only findings and pairwise ranking;

- `critic` receives candidates, critic `ContextPack`, evidence interpretation, and validation context, then returns report-only critique findings and observations;

- `anotherAngle` receives the initial validation, editorial critique, dossier cards, and another-angle `ContextPack`, then returns one different contract-safe route;

- `writer` executes the alternative route into one challenger candidate;

- `finalGate` receives the delivered final candidate, `FinalQualityContract`, deterministic gate findings, validation context, evidence interpretation, ledger, rule registry, material plan, and attribution requirement resolution, then returns independent findings, observations, and final repair goals;

- `writer` may execute bounded final public-prose repair cycles after the revision loop if `finalQualityGate` returns `warning` or `critical`;

- `writer` receives the current best candidate plus validator repair goals, editorial goals, rejected moves, dossier/context pack, and constraints, then returns revised prose;

- directed revision attempts use `DRAFT_REVISION_TEMPERATURE` and `DRAFT_REVISION_TOP_P`; malformed-JSON repair and backup attempts use the JSON repair temperature;

- regression guard checks deterministic health, and `review` compares old-vs-new by idea strength, tension, reader value, author stance, source integration, structure, and validator health;

- acceptance policy decides whether the revised prose replaces the current best;

- `complete` receives only the final decision, not the whole model conversation.

Trace:

- `validation` step;
- child `AiRun.requestPayload.draftRunStep = llmValidation`;
- child `AiRun.requestPayload.draftRunStep = editorialCritique`;
- child `AiRun.requestPayload.draftRunStep = alternativeAngleRoute`;
- child `AiRun.requestPayload.draftRunStep = alternativeAngleCandidate`;
- child `AiRun.requestPayload.draftRunStep = pairwiseRanking`;
- child `AiRun.requestPayload.draftRunStep = directedRevision`.

4.12 `complete`

Purpose: mark the run result.

Output cases:

- normal success: parent DraftRun has `finalDraft`;
- quality-blocked success: parent DraftRun has `finalDraft = null` and `complete.status = blocked`;
- failure: parent DraftRun status is `failed` with a safe error.

Role/model handoff:

- no LLM role is used;
- this step only records the final orchestration outcome for the frontend and

trace workbench.

4.13 Post-run editor version loop

Purpose: let the human editor keep improving or manually editing the delivered machine draft without mutating older versions.

Inputs:

- parent `DraftRun.finalDraft`, stored locally as `DraftVersion v1` with

`source=machineFinal`;

- active draft version chosen by the editor;
- optional editor comment;
- optional compact machine trace context from the parent DraftRun: final quality

gate summary, revision loop summary, alternative-angle outcome, validation summary, unresolved risks, post contract, and material evidence.

Behavior:

- the editor can switch between saved versions;
- manual text edits are local until the user saves them as a new

`source=manualEdit` version;

- one click on `Улучшить по комментарию` calls

`POST /api/drafts/revise-with-comment`;

- that endpoint is not a new DraftRun. It is a post-run writer-role JSON call with

child `AiRun` audit and existing JSON retry discipline;

- after a successful writer revision, the endpoint runs a lightweight `review` role

quality check. It evaluates whether the revised version followed the editor comment, preserved visible source markers from the base version, avoided internal pipeline jargon, and did not regress as public prose;

- quality check failure or provider unavailability does not cancel the created

version. The version receives `qualityCheck.status = notRun` with attempt metadata;

- provider failure does not create a fake version;

- the editor can mark any saved version as final, including `v1` after later

versions exist;

- final version selection creates or updates one deterministic

`editorialLearning` note in Author Memory. The note is visible immediately with status `pendingReview`, but it does not influence author-position inference until the editor accepts it.

Output:

- local `PostDraft.versions[]`, `activeVersionId`, and `finalVersionId`;

- compatibility mirror fields `PostDraft.title/body/version` always reflect the

active version;

- `FinalText.draftVersionId`, `FinalText.versionNumber`, and

`FinalText.editorDecisionSnapshot`;

- `DraftVersion.qualityCheck` for human-comment revisions when the review ran or

explicitly could not run;

- one `AuthorNote.type = editorialLearning` with selected/rejected version metadata,

comment summaries, manual edit count, quality-check summaries, unresolved risks, suggested takeaway, and status `pendingReview | accepted | rejected`.

Role/model handoff:

- post-run human-comment revision uses the `writer` role;

- it receives a compact machine context, not raw DraftRun JSON;

- it must preserve source markers, allowed claims, forbidden moves, and final

quality constraints.

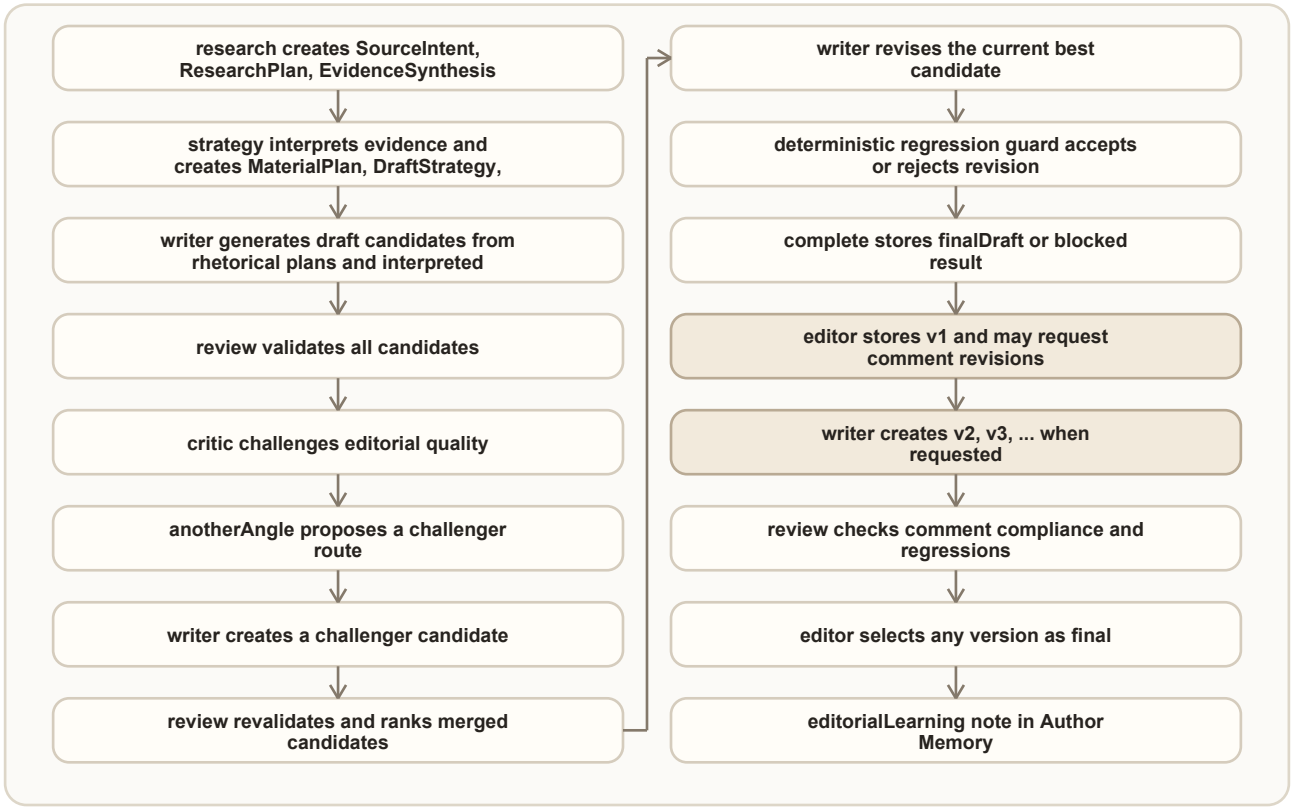
- post-run human revision quality check uses the `review` role;

- it is diagnostic: it can mark the version as passed, warning, critical, or not-run,

but it cannot block saving or final approval.

5. Role model usage and handoff AS IS

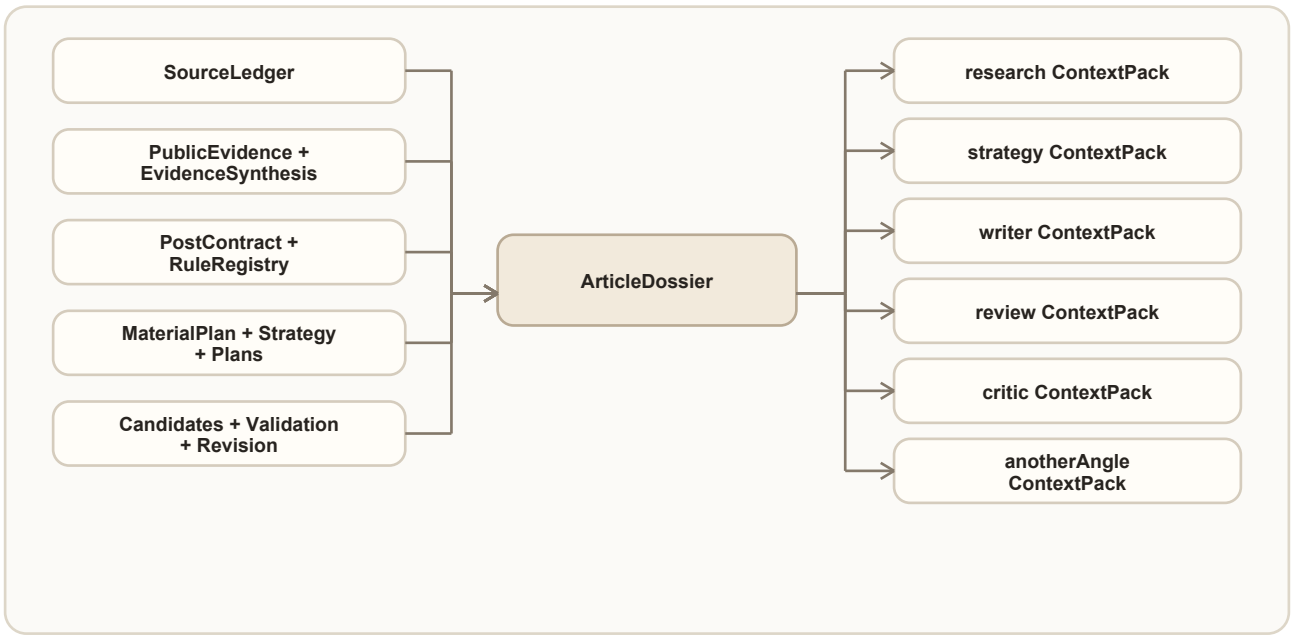
The current role interaction is artifact-mediated. A role does not continue the previous model's hidden conversation. It receives explicit artifacts and a role-specific context pack, then writes a new artifact for the next role.



Role summary:

Role	Used now	Receives	Must not do
`research`	source research planning and evidence synthesis	source intent, public evidence, context	invent facts or strengthen weak evidence
`strategy`	evidence interpretation, material plan, draft strategy, rhetorical plans	strategy context pack, rules, contract, material, evidence synthesis	change post thesis or bypass contract
`writer`	draft candidates and directed revisions	writer context pack, evidence interpretation, material plan, rhetorical plan, repair goals	create new claims or violate forbidden moves
`review`	LLM validation and pairwise ranking	review context pack, candidates, rules, ledger, validation reports	rewrite prose directly
`critic`	report-only editorial critique	critic context pack, evidence interpretation, candidates, validation context	rewrite prose or change final selection
`anotherAngle`	one critic-driven challenger route	another-angle context pack, initial validation, editorial critique, rejected/weak moves	act as technical backup or invent new evidence
`writer` post-run	human-comment revision of one saved version	active draft version, editor comment, compact machine trace context	mutate prior versions or fabricate a successful revision after provider failure
`review` post-run	comment-compliance quality check for human-comment revisions	base version, revised version, editor comment, compact machine trace context	block saving a successful revision or turn warnings into learning signals automatically

6. Context flow



Important AS IS rules:

- raw DraftRun trace is not the normal prompt context;
- `ArticleDossier` is rebuilt from selected artifacts;
- `ContextPack` selects up to 12 cards for one role;
- child `AiRun.requestPayload` stores the role model and context pack it received;
- context packs are local orchestration artifacts, not long-term memory.

7. Artifact catalog

Artifact	Created by	Read by	Debug location
`SourceLedger`	context/publicEvidence	feasibility, contract, rules, planning, validation	`context.sourceLedger`, `publicEvidence.enrichedSourceLedger`
`SourceIntent`	sourceIntent	publicEvidence	`sourceIntent` step
`ResearchPlan`	sourceIntent	publicEvidence	`sourceIntent.researchPlan`
`PublicEvidenceBatch`	publicEvidence	evidence synthesis, trace	`publicEvidence.items`, `publicEvidence.attempts`
`EvidenceSynthesis`	publicEvidence	ledger merger, material planning, trace	`publicEvidence.evidenceSynthesis`
`DraftRunBudget`	context	sourceIntent, publicEvidence, material plan, draft, validation	`context.draftRunBudget`, step budget metadata
`EvidenceInterpretation`	rulePack	material plan, writer, dossier, trace	`rulePack.evidenceInterpretation`
`PostContract`	postContract	rules, planning, validation, revision	`postContract` step
`RuleRegistrySnapshot`	rulePack	material plan, validation, revision	`rulePack.ruleRegistrySnapshot`
`MaterialPlan`	materialPlan	strategy, writer, validation, ranking	`materialPlan.materialPlan`
`DraftStrategy`	strategy	rhetorical plans, writer	`strategy.draftStrategy`
`RhetoricalPlanSet`	rhetoricalPlans	draft candidates	`rhetoricalPlans.rhetoricalPlanSet`

Artifact	Created by	Read by	Debug location
`DraftCandidate`	draft	validation, ranking, revision	`draft.candidates[]`
`ValidationReport`	validation	ranking/revision, trace	`validation.candidateReports`
`LlmValidationReport`	validation	ranking/revision, trace	`validation.llmValidationReport`
`EditorialCritiqueReport`	validation	alternative-angle tournament, trace, dossier	`validation.editorialCritiqueReport`
`AlternativeAngleTournament`	validation	final validation/ranking, trace	`validation.alternativeAngleTournament`
`RankingRevision`	validation	final draft selection, trace	`validation.rankingRevision`
`FinalQualityGate`	validation	final draft selection, trace	`validation.rankingRevision.finalQualityGate`
`ArticleDossier`	article memory service	context pack builder, trace	selected step artifacts
`ContextPack`	article memory service	child LLM services	step artifacts and child `AiRun.requestPayload`
`DraftVersion`	frontend/editor actions	editor version list, final selection	local workspace `postDraft.versions[]`
`HumanCommentRevisionQualityCheck`	post-run revise-with-comment endpoint	editor version list, final selection, future learning slice	local workspace `postDraft.versions[].qualityCheck`, child `AiRun.requestPayload.draftRunStep = humanCommentRevisionQualityCheck`
`EditorDecisionSnapshot`	final text approval	editorial-learning note builder, audit/debug	local workspace `finalText.editorDecisionSnapshot`
`EditorialLearningAuthorNote`	final text approval	author memory feed, optional author-position inference after acceptance	local workspace `authorNotes[]`, type `editorialLearning`

8. Retry, fallback, and blocked behavior

Area	Behavior
live DraftRun	frontend keeps polling; no compatibility fallback while run is alive
stale DraftRun	diagnostic state after no timestamp progress; not automatic fallback
slow-but-healthy validation	validation has `runtimeBudget.currentOperationId` and heartbeat/current-operation age inside `runtimeBudget.limits.staleAfterSeconds`; do not classify as stuck
over-budget validation operation	stale diagnostics report the current validation operation and runtime stale budget instead of the old generic five-minute message
public search disabled	attempt is `notConfigured`; it is not proof
malformed JSON planning	repair retry, optional backup model, then deterministic fallback
malformed, empty, or timed-out evidence interpretation	failed child `AiRun`, failed nested operation, repair retry, optional backup model, then deterministic interpretation fallback
editorial critique provider missing	`editorialCritiqueReport.status=not-run`; no fake critique
malformed editorial critique JSON	repair retry, optional backup model, then `not-run` for that candidate
alternative-angle provider missing	`alternativeAngleTournament.status=not-run`; original candidates continue
malformed alternative-angle route JSON	repair retry, optional backup model, then tournament `failed`; no deterministic fake angle
alternative-angle candidate provider failure	repair retry, optional backup model, then tournament `failed`; original candidates continue
validation sub-operation failure	mark nested operation failed, preserve partial artifact, keep previous best if available

Area	Behavior
material plan ignores evidence	accountability retry, optional backup model, then emergency fallback
unresolved attribution requirement	record diagnostic metadata; not a repair goal unless it resolves to expected source markers that are missing from the final body
human-comment revision provider failure	no new version is created; UI keeps the current version list and shows a safe error
human revision quality-check failure	created version is kept; `qualityCheck.status=notRun` records attempts and safe reason
candidate provider failure	fallback only for that candidate direction
fallback candidate selection	fallback is diagnostic unless publishability guard allows it
feasibility blocked	run succeeds with `finalDraft = null`; no local fallback
all invalid candidates	run succeeds as quality-blocked; no local fallback
revision regression	keep previous best candidate

Slice 2.17.4.6.0.3.2 adds shared `operationEnvelope` governance for representative provider-heavy operations. Migrated operations expose `operationId`, operation kind, owner, status, attempts, child `AiRun` ids, input/payload stats, retry policy, timeout profile where applicable, incident metadata, safe error, and result payload. Fallback, not-run, failed, timeout, cancelled, and stale outcomes require incident metadata. Backup success keeps the accepted payload but records `backupAccepted`. Representative migrated operations are `evidenceInterpretation`, `editorialCritique`, `directedRevision`, `humanCommentRevision`, and `humanCommentRevisionQualityCheck`; other current provider-heavy operations remain in `CURRENT_LLM_OPERATION_INVENTORY` until their owning migration slices.

Slice 2.17.4.6.0.3.3 adds DraftRun provider-input payload budget governance. Full artifacts stay in parent DraftRun storage, but enforced provider calls receive a curated compact payload immediately before prompt messages are built. The enforced representative operations are `evidenceInterpretation`, `editorialCritique`, `directedRevision`, `humanCommentRevision`, and `humanCommentRevisionQualityCheck`. Each records `payloadBudget.profileId`, execution mode, limits, sent/trimmed counts, suppressed fields, semantic input contract, `qualityRisk`, `promptCharEstimate`, and `approxTokenEstimate` in child `AiRun.requestPayload`, attempts, and `operationEnvelope.payloadStats`. If compacted input still exceeds the profile cap, trace records `contextOverBudget`; hard-cap breaches record `payloadTooLarge`. Remaining provider-heavy operations are explicit payload-budget debt entries in `CURRENT_LLM_OPERATION_INVENTORY`.

Slice 2.17.4.6.0.3.4 adds `ValidationRuntimeBudget` governance for the validation stage. The runtime guard records `runtimeBudget.profileId`, execution mode, limits, used counters, `startedAt`, `lastHeartbeatAt`, `currentOperationId`, `currentOperationStartedAt`, `slowButHealthy`, `stopReason`, `exhausted`, and incidents in the existing validation progress artifact. It uses canonical stop reasons: `acceptedQuality`, `humanReviewRequired`, `budgetExhausted`, `maxIterations`, `noImprovement`, and `providerIncident`; older internal reasons are kept as `detailStopReason` where useful. No endpoint contract or SQLite schema is changed.

9. How to read a run trace

Open `/ai-runs?runId=<DraftRun ID>` and inspect in this order:

1. `context`: confirm source signal, candidate, topic, fabula, rules are present.
2. `DraftRun budget`: confirm research depth, execution mode, effective caps, used

counts, cap hits, skipped tasks, and trimmed evidence/claims.

3. `sourceIntent`: confirm research requests were normalized correctly.
4. `publicEvidence`: confirm URL/search attempts, accepted evidence, rejected drift,

evidence synthesis, and enriched ledger.

5. `feasibility`: confirm run was allowed or blocked for quality reasons.

6. `postContract`: confirm thesis, CTA, size, allowed claims, and forbidden moves.
7. `rulePack`: confirm rule registry exists and size/evidence rules are present.
8. `evidenceInterpretation`: inside `rulePack`, confirm implications, examples,

limits, forbidden overclaims, rejected evidence uses, attempts, `operationEnvelope`, timeout profile, input stats, `payloadBudget`, and incident taxonomy.

9. `materialPlan`: confirm selected evidence and explicit rejection reasons.
10. `strategy`: confirm the strategy does not change the contract.
11. `rhetoricalPlans`: confirm the plans are different routes, not new topics.
12. `draft`: compare candidates, source/fallback status, publishability, scorecard.
13. `validation`: inspect deterministic warnings, LLM findings, observations.
14. `editorialCritiqueReport`: inspect idea strength, tension, author stance, source

integration, generic-prose risks, recommended editorial moves, candidate-level `operationEnvelope`, malformed JSON/schema incidents, and not-run incidents.

15. `alternativeAngleTournament`: inspect challenger route, candidate, model attempts,

and why it entered or failed to enter final ranking.

16. `rankingRevision`: inspect pairwise winner, revision cycles, accepted/rejected moves.

Directed revision sub-results should include `operationEnvelope` for accepted, failed, or not-run writer revisions. `revisionLoop.stopReason` is canonical; inspect `detailStopReason` for legacy detail such as `editorially-improved`, `no-fresh-angle`, or provider failure detail. Inspect `runtimeBudget` to see max wall-clock, LLM calls, revision cycles, pairwise rounds, final repair cycles, non-improvement count, heartbeat, and slow-but-healthy status.

17. `finalQualityGate`: inspect the final quality contract, independent review

attempts/model, final public-prose status, `attributionReview`, `actionableAttributionFindings`, `diagnosticAttributionNoise`, repair goals, accepted or rejected repair cycles, and final draft source.

18. `validation.progress`: inspect nested operations; failed late operations should show

safe errors and the final previous-best decision when available.

19. child `AiRun` detail: inspect prompt messages, role model, generation params,

context pack, provider result, and sanitized raw response excerpt on JSON failures.

20. post-run editor state: inspect local `postDraft.versions[]` and

`finalText.editorDecisionSnapshot` to see which version the human selected, which comments created new versions, and whether machine trace context was available when the final text was approved.

21. post-run quality check: for human-comment revisions, inspect

`postDraft.versions[].qualityCheck` and child `AiRun.requestPayload.draftRunStep = humanCommentRevisionQualityCheck` to confirm matched/missed comment intents, source-marker preservation, public-prose status, internal jargon leaks, review attempts, incident metadata, and nested `operationEnvelope` on the revision/quality attempts.

22. post-run learning note: inspect `authorNotes[]` for type `editorialLearning`.

Pending and rejected notes should be visible but not used as author-position evidence. Accepted notes should flow through normal author-memory event and inference logic.

23. incident blast radius: when the same `incidentType` repeats across a run, compare

it against `CURRENT_LLM_OPERATION_INVENTORY` to identify other migrated or allowlisted operations with the same expected incident coverage before deciding whether this is an isolated provider error or a systemic architecture issue.

24. payload budget blast radius: when `qualityRisk`, `contextOverBudget`, or

`payloadTooLarge` repeats, compare `payloadBudget.profileId`, sent/trimmed counts, and inventory `payloadBudgetStatus` before blaming prompt quality.

10. Known AS IS limitations

- Editorial critique is active and feeds one alternative-angle challenger plus revision goals, but it is not yet an autonomous multi-agent debate loop.
- `anotherAngle` is active as one challenger route only, not as an autonomous debate loop or general fallback.
- `ArticleDossier` is a local compact memory artifact, not a vector store or cross-run RAG system.
- Failed late provider operations are safely finalized when the Python call returns or raises. Evidence interpretation now has an operation-level timeout envelope; a broader infrastructure watchdog is still needed for externally stuck Celery tasks outside protected operation envelopes.
- The pipeline has detailed traceability. The main editor UI receives one machine draft first, then manages post-run human versions locally. Cross-post learning is currently limited to reviewable `editorialLearning` author-memory notes; promotion into actual editorial rules remains a future slice.

11. Maintenance rules

- This Markdown document and the PDF must be updated together.
- Regenerate the PDF after every edit:

```
python scripts/generate-draft-run-pipeline-pdf.py
```

- Future DraftRun planning, implementation, diagnostics, evaluation, and autofix work must read this document before changing or judging the pipeline.
- If current behavior differs from this document, update the document first or include the update in the same slice as the behavior change.